Echolocating bats



Technological advances have greatly helped research into the hunting strategies of animals. Insectivorous bats (also known as microbats) use echolocation in the dark to locate and pursue flying insect prey, such as moths and midges. A team of Japanese scientists set out to study the bats' feeding behavior using microphone array measurements and mathematical modeling analysis. The researchers recorded the echolocation calls and flight paths of the bats and discovered that they often directed their sonar not just at their immediate prey but at the next target they were lining up as well.

The team also found evidence that the bats chose flight paths that would allow them to plan two steps ahead, rather like skilled chess players. Not only were the animals maximizing their energy input by targeting multiple prey items, but they were also minimizing their energy output by reducing the distance they flew in pursuit of insects. This behavior fits in well with optimal foraging theory.

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